

Kazi Md. Tawsif Rahman

Java Backend Engineer

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linkedin.com/in/tawsifrahman113 | github.com/tawsif113 | Engineering Portfolio

Java backend engineer building CRM, ticketing, and lending services with Spring Boot, PostgreSQL, Redis, RabbitMQ, and MongoDB. Experience with configurable business rules, workflow automation, and DDD/CQRS; independent projects demonstrate concurrency control, reliable messaging, security, and integration testing.

Professional Experience

Software Engineer

Dec 2024 – Present

BRAC IT Services Ltd. | Dhaka, Bangladesh

CRM Platform

- Developed Spring Boot services for lead-to-quote workflows and sales pipelines; implemented configurable ownership assignment based on priority, schedules, conditions, and rotation rules.
- Optimized Kanban ordering by replacing full-column reordering with fixed-length order keys, reducing routine card moves to a single rank assignment.
- Implemented idempotent card moves, board access controls, and audit trails; synchronized opportunity columns with pipeline stages and validated allowed transitions before changing business state.

Ticketing System

- Developed ticket lifecycle services covering creation, search, comments, forwarding, assignment, and notifications using Spring Boot, PostgreSQL, Redis, and RabbitMQ.
- Automated least-loaded ticket assignment using Redis-backed workload counters and a min-heap, incorporating resolver availability and reliever fallback to reduce manual routing.
- Implemented SLA and KPI tracking with assignee-restricted stage updates, progress notifications, and activity histories; maintained workload counters through asynchronous updates following ticket creation, reassignment, and resolution.

Loan Proposal Command Service

- Developed command workflows for loan proposal creation, updates, bulk processing, approval, and soft deletion in a DDD/CQRS microservice using Spring Boot, MongoDB, and RabbitMQ.
- Encapsulated eligibility, loan limits, installment, insurance, and disbursement rules in a loan-proposal aggregate and composable business specifications; added aggregate-level tests for proposal updates.
- Implemented atomic monthly proposal-number generation, batch duplicate checks, and lifecycle domain events to support proposal identification, validation, and downstream integration.

Education

B.Sc. in Computer Science and Engineering

Sep 2024

Chittagong University of Engineering and Technology (CUET), Bangladesh

CGPA: **3.59 / 4.00**

Selected Independent Projects

Spring Boot Rescue Lab

GitHub

Java 25, Spring Boot 4, PostgreSQL, Redis, RabbitMQ, Testcontainers, GitHub Actions

- Built a reproducible lab for six simulated backend incidents covering SQL performance, duplicate requests, inventory races, authorization, event delivery, and cache stampedes; documented fixes, trade-offs, and regression evidence.
- Reduced SQL statements from **42 to 3** for a 20-order page using pagination-safe fetching. Collapsed **24 concurrent cold reads to 1 database load** with Redis single-flight coordination and TTL jitter in controlled integration tests.
- Implemented persistent idempotency with request fingerprints and PostgreSQL advisory locks; verified that **8 concurrent retries create 1 order** and reserve stock once. Used atomic conditional updates to prevent inventory overselling.
- Added a transactional outbox, RabbitMQ publisher confirms, scheduled retries, dead-letter queues, and consumer deduplication; validated database atomicity and retry bookkeeping with failure injection, and automated checks through GitHub Actions.

Enterprise IAM Lab

GitHub

Keycloak, Spring Security, OAuth 2.0 / OIDC, PKCE, OpenLDAP, Docker Compose

- Built an identity integration lab with two OIDC clients to explore Authorization Code + PKCE, shared SSO sessions, LDAP user federation, and role/group mapping.
- Configured a Spring Boot resource server with JWT issuer and audience validation, Keycloak role-to-authority conversion, and method-level authorization for employee, support-agent, and administrator endpoints.

Technical Skills

Languages	Java, SQL
Backend	Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST APIs, OpenAPI, MapStruct
Data & Messaging	PostgreSQL, MongoDB, Redis, RabbitMQ, Flyway
Architecture	DDD, CQRS, event-driven systems, transactional outbox, idempotency, concurrency control
Identity & Security	Keycloak, OAuth 2.0, OpenID Connect, PKCE, JWT, role-based access control
Testing & Delivery	JUnit 5, Mockito, Testcontainers, GitHub Actions, Gradle, Docker, Docker Compose, Git, Linux
Observability	Spring Boot Actuator, Micrometer, Prometheus, Grafana, structured logging

Achievements

Codeforces Specialist (1500+ rating); finalist at CUET IUPC 2023 & 2024 and CUET Tech Carnival 2020.

Selected Publication

An Automated System for Detecting Property Insurance Fraud Using Machine Learning

Kazi Md. Tawsif Rahman, Chowdhury Mahfuzul Hoq. *IJMSC*, Sep 2024.

DOI: 10.5815/ijmsc.2024.03.02